



DISTRICT-LEVEL DENGUE OUTBREAK EARLY WARNING SYSTEM FOR SRI LANKA

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Declaration

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We further affirm that artificial intelligence-based tools were used only in a limited, responsible, and transparent manner during the preparation of this intermediate. These tools were employed solely for language refinement, structural suggestions, and preliminary idea exploration. All substantive content, including the conceptualization, analysis, interpretation of results, and conclusions, is entirely our own. We take full responsibility for the accuracy, originality, and integrity of the work presented herein.

Research Name: District-Level Dengue Outbreak Early Warning System for Sri Lanka

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Abstract

Dengue remains a recurring public-health challenge in Sri Lanka, where disease incidence varies considerably across time, geographical areas, climatic zones, and monsoon periods. Existing surveillance is primarily retrospective, while many previous forecasting studies are geographically limited, use monthly aggregation, or evaluate individual modelling approaches. This research investigates the development of a district-level dengue outbreak early-warning framework capable of supporting localised four-week-ahead prediction.

The study constructs a longitudinal weekly dataset covering January 2016 to 31 May 2026 across all 26 Regional Director of Health Services administrative areas represented in Sri Lanka's Weekly Epidemiological Reports. Historical weekly dengue case counts are integrated with precipitation, temperature, and relative-humidity observations obtained through the NASA POWER API. The data are standardised into a common geographical and weekly structure for exploratory analysis, climate-lag investigation, feature engineering, and predictive modelling.

The proposed methodology is designed to compare SARIMAX, LightGBM, and Long Short-Term Memory models under a common four-week-ahead, time-aware validation framework. The system is designed to generate both numerical dengue case forecasts and localised outbreak-risk alerts for presentation through an interactive decision-support dashboard.

At the current intermediate stage, data acquisition, preprocessing, spatiotemporal integration, and exploratory analysis have been completed. The results demonstrate substantial temporal variation in dengue incidence and clear differences in cumulative disease burden between climatic zones, with approximately 67.5% of the observed cumulative cases occurring in the Wet Zone. National climate visualisations do not show a simple same-week relationship between meteorological variables and major dengue peaks, supporting further investigation of delayed and localised climatic effects.

The comparative forecasting, outbreak-classification, and complete dashboard evaluation stages remain to be completed before final predictive-performance conclusions can be established.

Keywords: Dengue Forecasting, Early Warning System, Climate-Lag Analysis, SARIMAX, LightGBM, LSTM, RDHS, Sri Lanka

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List of Abbreviations

API	Application Programming Interface
ARIMA	Autoregressive Integrated Moving Average
ARIMAX	Autoregressive Integrated Moving Average with Exogenous Variables
CNN	Convolutional Neural Network
GIS	Geographic Information System
LightGBM	Light Gradient Boosting Machine
LSTM	Long Short-Term Memory
MAE	Mean Absolute Error
MAPE	Mean Absolute Percentage Error
NASA POWER	National Aeronautics and Space Administration Prediction Of Worldwide Energy Resources
RDHS	Regional Director of Health Services
RMSE	Root Mean Squared Error
ROC-AUC	Area Under the Receiver Operating Characteristic Curve
SARIMAX	Seasonal Autoregressive Integrated Moving Average with Exogenous Variables
SHAP	SHapley Additive exPlanations
XGBoost	Extreme Gradient Boosting

Chapter 1

Introduction

1.1 Background

Dengue fever is a major mosquito-borne public-health problem in Sri Lanka and has remained a recurring national health burden for many years [1]. Dengue transmission is affected by the interaction of epidemiological, environmental, seasonal, and geographical factors. In particular, precipitation, temperature, and humidity can influence mosquito breeding, survival, and disease transmission. Sri Lanka's monsoonal climate further contributes to strong seasonal and regional variations in dengue incidence [2, 3].

Previous studies have shown that dengue behaviour is not uniform across the country. Different districts and climatic zones can experience different outbreak patterns and seasonal peaks. Prabodanie et al. demonstrated clear spatiotemporal differences in dengue outbreaks across Sri Lanka, including differences between wet-zone and dry-zone areas [2]. Similarly, studies on the relationship between climate and dengue have shown that changes in weather conditions may affect dengue incidence only after a time delay, making lagged climatic variables important for early-warning applications [3].

Despite the availability of national surveillance data, dengue-control activities are still largely dependent on identifying increases after cases have already been reported. Such retrospective surveillance is essential for monitoring the disease, but it provides limited lead time for preventive action. A forecasting system that can estimate dengue risk several weeks in advance could support earlier vector-control activities, preparedness, and resource allocation.

Recent research has increasingly explored statistical, machine-learning, and deep-learning techniques for dengue prediction. Sri Lankan studies have demonstrated the usefulness of time-series models and weather-based predictors, while also showing that model performance and weather relationships vary across locations [4]. However, much of the existing work is limited to selected districts, monthly aggregation, a single modelling approach, or research-focused prediction experiments without an integrated decision-support interface. Reviews of Sri Lankan dengue research also indicate that many areas remain less represented than frequently studied regions such as Colombo and Gampaha [5].

To address these limitations, this research develops a district-level dengue outbreak early-warning system using a unified weekly spatiotemporal dataset for all 26 Regional

Director of Health Services (RDHS) administrative health areas represented in the national Weekly Epidemiological Reports. Historical weekly dengue cases are integrated with precipitation, temperature, and relative-humidity data obtained through the NASA POWER API. The research is designed to compare SARIMAX, LightGBM, and Long Short-Term Memory (LSTM) models under a common forecasting framework and to investigate district/RDHS-specific climate-to-dengue time lags. The intended output is a four-week-ahead dual-stream prediction consisting of a numerical dengue case forecast and a localised outbreak-risk alert, presented through an interactive decision-support dashboard.

1.2 Problem Statement

Sri Lanka has an established dengue surveillance process, but reported case information mainly describes infections that have already occurred. Consequently, public-health interventions may be initiated after transmission has already increased. Effective outbreak prevention requires a practical mechanism that can convert historical epidemiological and climatic data into reliable, location-specific warnings before a surge becomes severe.

Several limitations in existing dengue forecasting research make this difficult:

1. **Limited geographical generalisation:** Many existing Sri Lankan studies focus on one or a small number of districts. Dengue behaviour differs across districts and climatic regions, while the existing spatial literature is geographically uneven. Therefore, findings or models developed for a single location should not be assumed to generalise reliably to the entire country [2, 5].
2. **Insufficient use of delayed climate effects:** Precipitation, temperature, and humidity do not necessarily influence reported dengue cases during the same time period. Biological and environmental processes can create delayed climate–dengue relationships, and Sri Lankan studies show that time-shifted weather variables can provide stronger relationships with dengue incidence. These delayed relationships should therefore be investigated at local level rather than assuming a same-week effect [3, 4].
3. **Comparative forecasting gap:** Statistical, machine-learning, and deep-learning models have different strengths. Existing Sri Lankan comparative work has been conducted using selected geographical areas and does not establish which approach performs most reliably across all 26 RDHS administrative health areas under the same weekly dataset, forecasting horizon, and validation procedure [4, 5].
4. **Operational decision-support requirement:** A high-performing forecasting model

alone is not sufficient for practical public-health use. Forecast outputs must be translated into understandable case estimates, localised risk alerts, temporal trends, and visualisations that can support timely interpretation and decision making by health authorities.

Taken together, these limitations indicate a gap in nationally scoped weekly dengue early-warning research at RDHS level in Sri Lanka [3–5]. Such a framework should combine historical case patterns with lagged climatic information, compare multiple forecasting paradigms, and convert the resulting predictions into practical four-week-ahead decision-support outputs.

1.3 Research Objectives

The primary objective of this research is to design and develop a proactive, data-driven dengue outbreak early-warning and decision-support system for Sri Lanka that can provide district/RDHS-level forecasts four weeks in advance.

The specific objectives are:

1. **Data Fusion:** To integrate historical weekly dengue cases extracted from the Epidemiology Unit’s Weekly Epidemiological Reports with synchronised meteorological data, including precipitation, temperature, and relative humidity from the NASA POWER API, into a unified longitudinal panel dataset covering all 26 RDHS administrative health areas.
2. **Comparative Modelling Framework:** To implement and evaluate SARIMAX, LightGBM, and LSTM forecasting pipelines under a common predictive and validation framework in order to compare statistical, machine-learning, and deep-learning approaches.
3. **Dual-Stream Prediction Output:** To generate both continuous four-week-ahead dengue case-count forecasts through regression and localised outbreak-risk alerts through classification.
4. **Regional Adaptation and Lag Analysis:** To identify RDHS-specific climate-to-dengue time lags and capture variations associated with Sri Lanka’s Wet, Dry, and Intermediate climatic zones and monsoonal patterns.
5. **National-Scale Generalisation:** To extend the predictive framework across all 26 RDHS administrative health areas rather than limiting the analysis to a small number of frequently studied locations.
6. **Decision-Support Interface:** To develop an interactive web-based dashboard that presents forecasts, alert levels, temporal trends, and geographical risk information in a form suitable for public-health decision support.

1.4 Research Questions

This research addresses the following questions:

1. **RQ1:** What temporal relationships and district/RDHS-specific time lags exist between weekly dengue incidence and climatic variables such as precipitation, temperature, and relative humidity across Sri Lanka's different climatic zones?
2. **RQ2:** Which of SARIMAX, LightGBM, and LSTM provides the most reliable four-week-ahead dengue case forecasts across the 26 RDHS administrative health areas when evaluated under a common validation framework?
3. **RQ3:** How effectively can the selected predictive framework identify localised outbreak or high-risk conditions four weeks in advance using a classification-based alert mechanism in addition to numerical case forecasting?
4. **RQ4:** How can the resulting district/RDHS-level forecasts and risk alerts be presented through an interactive dashboard to provide understandable and actionable decision support for public-health authorities?

1.5 Scope and Limitations

1.5.1 Scope

The scope of this research is defined as follows:

- **Geographical scope:** The study covers all 26 RDHS administrative health areas listed in Sri Lanka's Weekly Epidemiological Reports. This enables RDHS-level analysis rather than limiting the work to a small set of regions.
- **Temporal scope:** The longitudinal dataset covers weekly observations from January 2016 to 31 May 2026, subject to the availability and consistency of the source records used in the study.
- **Epidemiological data:** Weekly dengue case counts are obtained from the Ministry of Health Epidemiology Unit's Weekly Epidemiological Reports and organised using year, epidemiological week, and RDHS area.
- **Climatic data:** Precipitation, temperature, and relative-humidity variables are obtained through the NASA POWER API and aggregated into weekly values aligned with the epidemiological records.
- **Analytical scope:** The study investigates historical dengue patterns, climatic-zone differences, monsoonal behaviour, lagged climate effects, and engineered temporal features for four-week-ahead prediction.

- **Modelling scope:** Three representative approaches will be compared: SARI-MAX as a statistical time-series baseline, LightGBM as a machine-learning model for structured lag-based features, and LSTM as a deep-learning sequence model.
- **Prediction scope:** The proposed system is designed to produce two complementary outputs: predicted dengue case counts and localised outbreak-risk alerts for a four-week-ahead forecasting horizon.
- **System scope:** The research includes the planned development of a web-based dashboard prototype for presenting forecasts, alerts, trends, and geographical risk information to support public-health decision making.

1.5.2 Limitations

The study has the following limitations:

- **Human-mobility data exclusion:** Human mobility was included in the initial research plan, but access to telecommunications-based mobility data was not possible because of privacy, security, and institutional data-governance restrictions. The variable was therefore formally removed from the implemented research scope.
- **Meteorological data source:** Continuous official district-level historical weather data from the Sri Lanka Department of Meteorology could not be acquired within the available undergraduate research budget. NASA POWER data are therefore used as a consistent alternative climatic source and may not capture every local micro-climatic variation observable at ground weather stations.
- **Historical surveillance-data structure:** Weekly Epidemiological Reports were published as PDF documents with formatting changes across years. A substantial part of the historical dengue dataset therefore required careful manual extraction and validation, which is more time-consuming and may remain susceptible to human transcription error despite verification procedures.
- **Predictor coverage:** The models focus primarily on historical dengue incidence and climatic variables. Other relevant determinants, including detailed entomological indices, human mobility, local vector-control interventions, and socio-economic conditions, are not comprehensively represented.
- **Prototype status:** The dashboard is developed as a research decision-support prototype and is not intended to replace official surveillance systems, epidemiological assessment, or clinical/public-health judgement without further operational validation.

1.6 Significance of the Study

This research is significant because it attempts to move dengue surveillance from retrospective monitoring toward proactive, data-driven early warning. Its main contributions are:

- **Broader geographical coverage:** The study develops a common forecasting framework for all 26 RDHS administrative health areas, reducing the geographical concentration found in many previous Sri Lankan dengue studies.
- **District/RDHS-specific climate-lag analysis:** By investigating delayed relationships between climate and dengue at local level, the study aims to reflect regional and monsoonal differences rather than assuming a single national pattern.
- **Model comparison:** The three modelling approaches will be evaluated using the same forecasting horizon, dataset structure, and validation procedure. This is intended to provide evidence for selecting an appropriate model for the Sri Lankan context.
- **Practical four-week-ahead output:** Producing both numerical case forecasts and outbreak-risk alerts is intended to provide a more actionable form of early warning than a single model-accuracy result.
- **Decision-support implementation:** The proposed dashboard is designed to translate model outputs into understandable visual information that can support prioritisation, preparedness, and resource-planning activities by public-health authorities.
- **Reusable research foundation:** The integrated weekly epidemiological and climatic dataset, together with the planned feature-engineering, model-comparison, and evaluation workflow, provides a foundation for future extensions using additional data sources and operational validation.

1.7 Intermediate Organisation

This intermediate is organised into five chapters.

Chapter 1: Introduction presents the research background, problem statement, objectives, research questions, scope and limitations, significance of the study, and overall Intermediate Organisation.

Chapter 2: Literature Review examines previous research on dengue epidemiology, spatiotemporal patterns, and climate–dengue relationships. It also reviews forecasting methods, machine-learning and deep-learning approaches, early-warning systems, and the gaps addressed by this study.

Chapter 3: Methodology describes the research design, data sources, data collection and preprocessing procedures, spatiotemporal data integration, exploratory and lag analysis, feature engineering, comparative modelling framework, validation strategy, evaluation metrics, and dashboard development process.

Chapter 4: Results and Analysis presents the results available at the intermediate stage of the research, including national dengue time-series patterns, climatic-zone differences, national weekly climate–dengue visualisations, and the current status of the research questions. It also identifies the climate-lag analysis, comparative model evaluation, outbreak-classification, and dashboard-validation results that remain to be completed.

Chapter 5: Conclusion summarises the major findings and contributions of the research, discusses the study limitations, and presents recommendations and directions for future work.

Chapter 2

Literature Review

2.1 Overview

Dengue forecasting and outbreak early warning require an understanding of several interconnected research areas, including dengue epidemiology, spatial and temporal disease patterns, climatic influences, forecasting models, outbreak-risk assessment, and decision-support systems. In Sri Lanka, dengue is endemic and recurrent, while outbreak timing and magnitude vary across districts and climatic regions [1, 2]. These differences make a single national prediction pattern insufficient for localised early warning.

The literature reviewed in this chapter is therefore organised around five major themes: dengue epidemiology and spatiotemporal patterns, climate–dengue relationships, dengue forecasting approaches, early warning and decision support, and evaluation methods and research gaps. Particular attention is given to Sri Lankan studies because they provide the strongest evidence regarding local seasonality, spatial differences, weather effects, and forecasting challenges. International studies are also reviewed where they provide methodological support for machine-learning, deep-learning, interpretability, and operational early-warning approaches.

Figure 2.1 summarises the main themes considered in this literature review and illustrates how the existing evidence leads toward the proposed district-level dengue outbreak early-warning system.

2.2 Theoretical Foundations

The theoretical foundation of this research is based on the understanding that dengue transmission is both temporally dynamic and spatially heterogeneous. Dengue incidence is influenced by historical disease patterns, seasonal conditions, climatic variables, geographical characteristics, and biological processes associated with mosquito breeding and disease transmission. Sri Lankan studies demonstrate substantial spatial, seasonal, and climate-related variation in dengue dynamics [2, 3]. These relationships provide the foundation for constructing a localised early-warning framework.

Dengue epidemiology in Sri Lanka. Dengue remains one of the most persistent

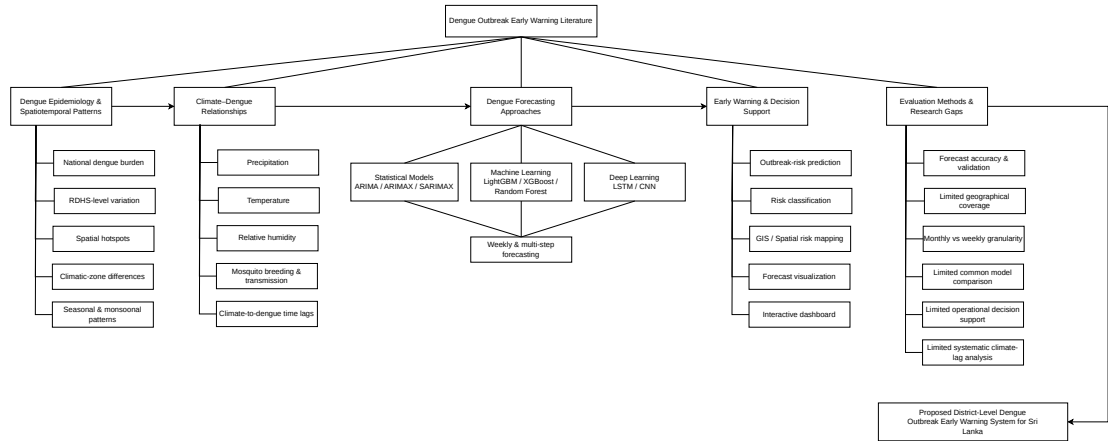


Figure 2.1: Taxonomy of the literature reviewed for dengue outbreak early warning, showing the main themes and their relationships.

vector-borne public-health problems in Sri Lanka. Fernando et al. reviewed the epidemiological burden of dengue in Sri Lanka from 2000 to 2020 and identified dengue as a continuing national health concern, including major high-burden periods such as 2017 and 2019 [1]. Such evidence demonstrates that dengue cannot be treated as an isolated or temporary outbreak problem. Instead, continuous surveillance and proactive outbreak preparedness are required.

However, national disease burden alone does not describe how dengue behaves within individual geographical areas. Spatial studies have demonstrated that dengue incidence is unevenly distributed. Edirisinghe used GIS-based analysis in Matara District and showed that dengue risk can concentrate within specific spatial and environmental settings rather than being uniformly distributed across an administrative area [6]. Sun et al. similarly investigated dengue and climate characteristics across Sri Lankan districts and identified geographically distinct hotspot and cold-spot patterns [7]. These findings indicate that localised modelling is more appropriate than assuming that every area follows the same national pattern.

Prabodanie et al. provide particularly strong evidence for regional variation in dengue dynamics. Their analysis of weekly dengue incidence across all 25 districts from January 2007 to July 2019 found that different climatic regions display different epidemic patterns [2]. Southwest wet-zone districts commonly experienced two dengue epidemic waves during a year, whereas northern and eastern dry-zone coastal districts often displayed one major epidemic wave. Stronger synchrony was also observed among several wet-zone districts. These findings support the inclusion of geographical and climatic-zone information in dengue forecasting.

A systematic review by Wickramasinghe et al. further highlighted the spatial imbalance in Sri Lankan dengue research [5]. Many previous investigations have focused on a small number of frequently studied locations, particularly Colombo and Gampaha,

while other areas remain comparatively underrepresented. The review also indicates that spatial patterns and the factors examined in dengue research vary across locations. Therefore, climate–dengue relationships should not automatically be assumed to be spatially constant across the country.

Climate–dengue relationships. Environmental conditions are important components of dengue transmission. Sri Lankan studies have reported associations between precipitation, temperature, relative humidity, and dengue incidence or forecasting performance [3, 4]. However, the relationship between climatic conditions and reported dengue incidence is not necessarily immediate.

Edirisinghe examined precipitation patterns in relation to dengue epidemics in Sri Lanka and reported an association between rainfall patterns and subsequent dengue epidemic behaviour [8]. Withanage et al. provided stronger direct evidence for delayed climatic effects in a forecasting study conducted for Gampaha District [3]. Their analysis evaluated multiple lag periods and incorporated previous dengue incidence together with lagged rainfall, rainy days, temperature, and relative-humidity variables.

The existence of such delays is biologically and epidemiologically plausible. Climatic conditions may first affect mosquito breeding and vector development before human infection, incubation, healthcare seeking, diagnosis, and official reporting occur. Withanage et al. discuss these delayed processes when interpreting the lagged weather variables used in their forecasting models [3]. Consequently, climatic information from earlier periods may be more informative for prediction than climatic conditions measured during the same reporting period. This provides the theoretical basis for climate-lag analysis in the present research.

Uduwanage et al. also observed that same-month weather relationships were relatively weak but became stronger after weather variables were shifted in time [4]. This finding further supports the use of lagged precipitation, temperature, and humidity features rather than relying only on simultaneous climate observations.

Seasonality and regional adaptation. Dengue forecasting must also account for recurring temporal patterns. Sri Lanka experiences seasonal rainfall associated with the Southwest monsoon, Northeast monsoon, and inter-monsoon periods. Since these climatic conditions vary geographically, the timing of dengue peaks also differs between regions. The findings of Prabodanie et al. demonstrate that wet-zone and dry-zone districts can display different epidemic-wave structures [2]. Therefore, seasonal features, climate-zone information, and localised lag structures are theoretically relevant for district/RDHS-level prediction.

Together, these studies establish three central foundations for this research: dengue is a recurring national public-health problem, dengue behaviour varies spatially and seasonally across Sri Lanka, and climatic effects can influence dengue incidence after delayed time periods. These foundations motivate the use of a weekly RDHS-level

spatiotemporal dataset and localised climate-lag analysis in the proposed early-warning framework.

2.3 Existing Approaches

Existing dengue research uses several methodological approaches. These can broadly be grouped into spatial and GIS-based analysis, statistical time-series forecasting, machine-learning models, deep-learning models, and risk-mapping or early-warning approaches. Each approach provides useful capabilities but also presents limitations when considered for a nationwide weekly early-warning system.

Spatial and GIS-based approaches. Spatial analysis is useful for identifying geographical concentrations of dengue and understanding whether risk is distributed uniformly or clustered within particular areas. Edirisinghe demonstrated the use of GIS for identifying spatially vulnerable dengue areas within Matara District [6]. Sun et al. identified broader hotspot and cold-spot patterns across Sri Lankan districts [7]. Prabodanie et al. extended this understanding through weekly spatiotemporal analysis and demonstrated different epidemic structures across climatic regions [2]. These approaches are valuable for identifying where dengue risk is concentrated, but they primarily characterize existing or historical patterns rather than providing a complete multi-week predictive early-warning system.

Statistical time-series approaches. Traditional statistical time-series models are widely used in dengue forecasting because dengue incidence often contains strong temporal dependence. Karasinghe et al. developed a modified ARIMA model for weekly dengue incidence forecasting in Colombo District [9]. Their final ARIMA-based model demonstrated stronger performance on the training data than on the later validation period, with increased errors during the COVID-19-affected period. The study confirms that weekly dengue forecasting is feasible in the Sri Lankan context while also illustrating that models fitted to historical case patterns can lose accuracy when external disruptions affect transmission or disease reporting.

ARIMAX extends ARIMA by incorporating external predictors. Sun et al. demonstrated the relevance of climatic information to dengue patterns, supporting the use of external environmental variables [7]. SARIMAX extends this concept further by representing both seasonal behaviour and exogenous predictors. Uduwanage et al. compared CNN, XGBoost, and SARIMAX for dengue prediction in selected Sri Lankan districts and reported SARIMAX as the strongest model among those evaluated based on RMSE [4]. Their findings provide direct support for selecting SARIMAX as the statistical baseline in the present study.

The main limitation of several existing Sri Lankan statistical studies is geographical or temporal restriction. Withanage et al. focused on Gampaha District and used

monthly-level modelling [3], while Karasinghe et al. focused on Colombo [9]. Although such studies provide important evidence, their results do not establish whether a single model configuration will perform consistently across all RDHS areas.

Machine-learning approaches. Machine-learning models can represent nonlinear interactions among multiple predictor variables. This is useful when dengue risk is influenced by combinations of case-history features, weather lags, seasonal indicators, and geographical variables. Uduwanage et al. evaluated XGBoost alongside CNN and SARIMAX in Sri Lanka [4], demonstrating the applicability of machine learning to dengue forecasting.

Rahman et al. developed an interpretable tree-based dengue early-warning model in Bangladesh using Random Forest, XGBoost, and LightGBM [10]. Their framework integrated dengue surveillance with climatic, sociodemographic, and landscape variables. LightGBM was identified as the optimal tree-based model, while SHAP analysis was used to investigate important predictors. This study provides methodological support for using LightGBM with structured multivariable dengue data and for incorporating model interpretability. In the present research, lag-based features will be derived separately from the local weekly RDHS-level histories.

However, models developed in another country cannot be transferred directly to Sri Lanka without local validation. Climatic conditions, dengue seasonality, surveillance systems, urban environments, and geographical patterns differ between countries. Therefore, LightGBM is included in the present research as a comparative machine-learning approach to be trained and evaluated using Sri Lankan RDHS-level data rather than by assuming that performance observed elsewhere will be reproduced locally.

Deep-learning approaches. Deep-learning models are increasingly used for sequential dengue forecasting. LSTM networks are designed to learn temporal dependencies from ordered sequences and are therefore relevant to weekly dengue and weather histories. Chen and Moraga developed an LSTM-based framework for weekly dengue forecasting and outbreak detection across selected Brazilian cities [11]. Their mobility-enhanced model integrated historical dengue cases, lagged climate variables, and human-mobility information and outperformed the evaluated baseline LSTM configurations. Their work therefore provides methodological support for treating dengue forecasting as a sequential learning problem.

Nevertheless, the Brazil study included mobility data, while such data is not available for the current Sri Lankan implementation. Therefore, the present research does not assume that LSTM will automatically outperform statistical or tree-based models. LSTM is instead included as a deep-learning comparison model to determine whether sequence learning from historical dengue and climate variables provides additional value under Sri Lankan conditions.

Other recent international studies also demonstrate the potential of climate-aware

machine-learning approaches. Zhang et al. examined multi-frequency climate–dengue relationships and identified frequency-specific and time-lagged climatic effects within an interpretable forecasting framework [12]. Al Bannoud et al. evaluated a broad set of machine-learning models under the exceptional 2024 dengue surge in São Paulo and showed the value of lagged predictors, virtual data augmentation, and boosting-based ensembles for improving robustness under extreme outbreak conditions [13]. These studies further support comparative modelling and explicit testing of robustness rather than dependence on a single algorithmic family.

The reviewed approaches differ in their strengths, limitations, and suitability for dengue early-warning applications.

Table 2.1: Comparison of representative dengue forecasting and early-warning approaches

Approach	Main Strength	Limitation Relevant to This Study	Representative References
Spatial / GIS analysis	Identifies geographical hotspots, regional variation, climatic-zone differences, and spatially vulnerable areas.	Mainly describes historical spatial patterns and does not by itself provide multi-week outbreak forecasting.	[2, 6, 7]
Statistical time-series models	Models temporal dependence, seasonality, historical case patterns, and, in ARIMAX/SARIMAX, external climatic variables.	Several Sri Lankan studies are limited to selected districts or monthly aggregation, reducing national weekly generalisability.	[3, 4, 9]
Machine-learning models	Captures nonlinear relationships among epidemiological, climatic, lagged, and engineered features; tree-based models can also support interpretability.	Performance depends on local data and feature design, and results obtained in other countries cannot be assumed to generalize directly to Sri Lanka.	[4, 10]
Deep-learning models	Learns complex sequential dependencies from historical dengue and climate time-series data and supports multi-step forecasting.	Requires greater modelling complexity and data preparation; performance advantages must be demonstrated rather than assumed.	[4, 11]
Early-warning and risk-mapping approaches	Converts surveillance or prediction outputs into risk information, spatial visualisation, and information that can support preventive action.	Integration of predictive modelling with practical operational decision-support remains limited in the Sri Lankan context.	[14, 15]

2.4 Technologies and Tools

The reviewed literature demonstrates that dengue early-warning research depends not only on prediction algorithms but also on technologies for spatial analysis, data integration, interpretation, visualisation, and risk communication.

Geographical Information Systems and spatial risk mapping. GIS-based approaches are useful for understanding geographical variation and presenting areas of increased dengue vulnerability. Edirisinghe applied GIS to identify spatial dengue vulnerability in Matara District [6], while Sun et al. demonstrated broader spatial clustering across Sri Lankan districts [7]. Del-Aguila-Mejía et al. combined system-dynamics modelling with risk mapping for dengue surveillance [14]. These studies demonstrate the importance of translating epidemiological information into location-based representations that can support targeted interventions.

For the present research, this literature supports the inclusion of geographical risk visualisation in the decision-support layer. The purpose is not only to generate numerical predictions but also to allow differences between RDHS areas to be interpreted spatially.

Model explainability. Machine-learning models can produce strong predictions while remaining difficult to interpret. In public-health applications, understanding why a model produces a particular risk estimate can increase the usefulness of the prediction. Rahman et al. used SHAP analysis to explain the contribution of predictor variables in their tree-based dengue early-warning framework [10]. This provides direct methodological support for the use of feature-importance or SHAP-based analysis where applicable, particularly for the LightGBM component of the present study.

Visualisation and decision-support interfaces. Forecast outputs have limited practical value if they cannot be interpreted by intended users. Rahulan et al., in a case study of dengue control in Gampaha District, identified weaknesses in the use of available data and its translation into evidence-based action [15]. This highlights the need for systems that present predictions in an understandable and operational form.

The proposed study therefore treats visualisation as part of the early-warning process rather than as a separate presentation task. Forecasted case counts, risk status, temporal trends, climate patterns, and geographical differences are intended to be integrated into an interactive dashboard. Detailed implementation technologies selected for data processing, modelling, and dashboard development are described in Chapter 3.

2.5 Evaluation Methods

Reliable evaluation is essential when comparing dengue forecasting models because a model that performs well on historical training data may not necessarily provide reliable future predictions. The reviewed studies demonstrate the importance of quantitative error measurement and evaluation on observations that were not used for model fitting.

Uduwanage et al. compared CNN, XGBoost, and SARIMAX using RMSE and reported better performance for SARIMAX among the evaluated approaches [4]. Their study illustrates the importance of applying a common quantitative metric when comparing models developed for the same forecasting task.

Karasinghe et al. also demonstrated the importance of validation beyond training performance [9]. Although their modified ARIMA approach performed strongly during model development, performance weakened during a validation period affected by the COVID-19 pandemic. This shows that historical fit alone is insufficient and that models must be evaluated using future or unseen time periods.

Evaluation is especially important for the present research because three fundamentally different approaches—SARIMAX, LightGBM, and LSTM—will be compared. A fair comparison requires the same underlying dataset, forecasting horizon, and time-ordered validation procedure for all models. Randomly mixing future and past observations would not represent the real behaviour of an early-warning system because the model must generate a future forecast using only information available up to the prediction date.

The proposed study will therefore use time-aware walk-forward validation. Regression performance will be assessed using forecasting-error measures, while the outbreak-alert stream will be evaluated using classification measures. The exact validation procedure and the regression and classification metrics used in this study are defined in Chapter 3. This separation ensures that the literature review establishes the rationale for rigorous evaluation, while the methodology chapter documents the precise implementation.

2.6 Related Work

Several studies are particularly relevant to the design of the proposed dengue early-warning system because they address one or more of its core components: national or district-level dengue variation, climate-lag effects, weekly forecasting, comparative modelling, outbreak detection, interpretability, and operational decision support.

Prabodanie et al. [2]. This study analysed weekly dengue incidence across all 25 districts of Sri Lanka from 2007 to 2019. The authors identified clear regional differ-

ences in epidemic timing, including two annual epidemic waves in many southwest wet-zone districts and one dominant wave in several northern and eastern dry-zone coastal districts. The major strength of the study is its national geographical coverage and use of weekly data. However, it primarily focuses on spatiotemporal pattern analysis rather than developing a predictive forecasting model. The present research extends this direction by using regional and climatic-zone differences as predictive features within a four-week-ahead early-warning framework.

Withanage et al. [3]. Withanage et al. developed a dengue forecasting model for Gampaha District using previous dengue incidence and lagged meteorological variables. Their work demonstrates the predictive importance of delayed precipitation, temperature, humidity, and rainy-day effects. The main limitation for the current research context is geographical and temporal scope, as the model was developed for one district using monthly-level data. Nevertheless, the study provides strong local support for climate-lag analysis.

Karasinghe et al. [9]. Karasinghe et al. developed a modified ARIMA approach for weekly dengue incidence forecasting in Colombo District. The study is important because it demonstrates the feasibility of weekly forecasting using Sri Lankan surveillance data. However, the approach is primarily dependent on historical dengue values and was developed for a single district. Its validation performance was also affected during the COVID-19 period. The present study therefore retains historical case information but extends the predictor space to include climatic variables and compares statistical forecasting with machine-learning and deep-learning approaches.

Uduwanage et al. [4]. Uduwanage et al. compared CNN, XGBoost, and SARIMAX for dengue outbreak prediction in selected Sri Lankan districts using climatic information. Their analysis found stronger climate relationships after time-shifting weather variables and reported SARIMAX as the strongest model among those compared based on RMSE. This study directly supports both the inclusion of lagged weather variables and the selection of SARIMAX as a statistical baseline. However, the study was limited by weather-data availability for selected areas, and weekly dengue records were converted to monthly summaries. The present research attempts to extend the comparison to all 26 RDHS administrative health areas using a unified weekly dataset.

Rahman et al. [10]. Rahman et al. developed a tree-based dengue early-warning framework in Bangladesh using Random Forest, XGBoost, and LightGBM. LightGBM achieved the strongest overall performance, and SHAP was used to interpret predictor importance. The work supports the suitability of LightGBM for structured data containing epidemiological, climatic, and contextual variables. However, the study was conducted outside Sri Lanka. Therefore, the present research adopts LightGBM as a methodological candidate but evaluates it using local Sri Lankan data and locally derived temporal features.

Chen and Moraga [11]. Chen and Moraga applied LSTM models to weekly dengue forecasting and outbreak detection in Brazil and investigated the value of combining historical cases with climate and mobility information. Their mobility-enhanced LSTM outperformed the evaluated baseline configurations, supporting the use of LSTM for sequential dengue forecasting. However, mobility information formed part of their predictor set, whereas the current study does not include human-mobility data because of availability and feasibility constraints. LSTM is therefore used in the present study to test sequence-learning performance using dengue and climatic histories only.

Rahulan et al. [15]. Rahulan et al. examined dengue outbreak control and prevention in Gampaha District and highlighted operational weaknesses associated with the use of existing data and the translation of available information into evidence-based action. This finding is directly relevant to the decision-support component of the present study. It indicates that research contribution should not end with model accuracy; predictions must also be communicated in a form that supports practical public-health interpretation and action.

International evidence further reinforces this direction. Del-Aguila-Mejía et al. demonstrated the role of system-based outbreak modelling and spatial risk mapping in public-health decision support [14], while Feng et al. emphasized surveillance, early warning, coordinated response, and rapid action as important components of dengue prevention [16]. Zhang et al. highlighted complex temporal and lagged climate–dengue relationships [12], and Al Bannoud et al. demonstrated the use of robust ensemble machine learning and lagged predictor configurations under exceptional outbreak conditions [13].

Taken together, the related studies support the major components of the present research individually. However, within the literature reviewed here, these components are generally investigated in separate studies or under different geographical and temporal settings. The reviewed evidence does not demonstrate a single Sri Lankan weekly RDHS-level framework that brings together climate-lag analysis, statistical forecasting, machine learning, deep learning, outbreak-risk classification, and dashboard-based decision support under one common evaluation design.

2.7 Research Gap

A critical synthesis of the reviewed literature reveals several gaps that motivate the present study.

1. **Limited geographical coverage.** Many Sri Lankan dengue studies focus on one or a small number of locations, particularly Colombo, Gampaha, Matara, or other selected districts [3, 5, 9]. Although these studies provide valuable local evidence,

they do not establish how the same predictive approach performs across all RDHS administrative health areas.

2. **Monthly rather than weekly temporal granularity.** Several studies aggregate dengue or climatic information to monthly values [3, 4]. Monthly forecasting can identify broad seasonal behaviour but provides less operational lead-time detail than weekly forecasting. A weekly framework is more appropriate for short-term four-week-ahead early warning.
3. **Weather-data availability and representativeness.** Previous Sri Lankan work has been constrained by the geographical availability and representativeness of meteorological information. Uduwanage et al. reported that weather data were available only for selected districts because of data-access and budget limitations, and they also noted that a single station may not fully represent weather variation within a district [4]. A consistently available climatic source is therefore important when constructing a national-scale dataset covering all RDHS areas.
4. **Limited systematic analysis of climate lags.** Existing studies demonstrate that delayed weather relationships can be relevant to dengue prediction [3, 4, 8]. However, systematic comparison of multiple climate-lag periods across all RDHS areas remains limited. It therefore remains to be established whether the most useful lag periods differ by RDHS area, climatic variable, climate zone, or season.
5. **Limited common comparison of modelling paradigms.** Statistical, machine-learning, and deep-learning approaches have each been applied to dengue forecasting, but the reviewed studies evaluate them under different locations, datasets, temporal resolutions, predictor sets, and forecasting designs [4, 9–11]. Consequently, reported performance values should not be treated as directly comparable. There remains a need to evaluate representative statistical, machine-learning, and deep-learning approaches using the same Sri Lankan weekly dataset, four-week forecasting horizon, and validation procedure.
6. **Limited integration with operational decision support.** Many studies end with forecasting results, error metrics, or risk identification. However, operational dengue management requires information to be translated into understandable alerts and visualisations. Rahulan et al. highlight weaknesses in the translation of available dengue information into evidence-based action [15]. This demonstrates the need to connect predictive modelling with a practical dashboard-based decision-support interface.

The present research is designed to address these gaps within a focused and feasible scope. It does not attempt to incorporate every possible dengue determinant. Human mobility, detailed entomological indices, sanitation conditions, socioeconomic

factors, waste-management information, and local vector-control activities may influence dengue transmission, but such information is not consistently available across the required historical and geographical coverage.

Instead, the research focuses on consistently obtainable weekly dengue incidence and climatic information. The framework covers all 26 RDHS administrative health areas and integrates weekly dengue records with precipitation, temperature, and relative humidity. The planned analytical stages will investigate climate lags from one to twelve weeks, consider climatic-zone and monsoon-related variation, and compare SARIMAX, LightGBM, and LSTM under a common four-week-ahead forecasting setup.

The proposed framework is also designed to produce two complementary outputs. The regression component will estimate future dengue case counts, while the classification component will identify potential outbreak or high-risk conditions. These outputs are intended to be presented through an interactive dashboard for public-health decision support.

The novelty of the study therefore lies primarily in the integration and localised application of existing methods rather than the invention of a new forecasting algorithm. Existing literature independently supports spatial analysis, climate-lag modelling, statistical forecasting, machine learning, deep learning, explainability, risk mapping, and early warning. The present research is designed to bring these evidence-supported components together within one reproducible Sri Lankan RDHS-level weekly framework.

2.8 Chapter Summary

This chapter reviewed the literature relevant to the development of a district-level dengue outbreak early-warning system for Sri Lanka. The reviewed studies establish dengue as a recurring national public-health problem and demonstrate substantial spatial, seasonal, and climatic variation across Sri Lanka [1, 2]. They also show that climatic variables can be associated with dengue incidence after delayed time periods, supporting the investigation of lagged climatic predictors [3, 4].

The review examined statistical, machine-learning, and deep-learning forecasting approaches. Sri Lankan studies provide support for weekly time-series forecasting, lagged weather variables, and SARIMAX-based prediction [4, 9], while international studies provide methodological support for LightGBM, LSTM, interpretability, and risk-oriented early-warning approaches [10, 11, 14]. Sri Lankan operational evidence also emphasizes the importance of translating available data into usable information for public-health decision making [15].

Several research gaps were identified, including limited national geographical coverage, the use of monthly rather than weekly data in several studies, weather-data availability constraints, limited systematic climate-lag analysis, limited direct comparison of

different modelling paradigms under a common framework, and insufficient integration of forecasting results with operational decision-support interfaces.

These gaps provide the motivation for the methodology presented in Chapter 3, which describes the unified weekly RDHS-level dataset and the planned climate-lag analysis, feature engineering, comparative evaluation of SARIMAX, LightGBM, and LSTM, four-week-ahead regression and outbreak classification, and development of the proposed decision-support dashboard.

Chapter 3

Methodology

3.1 Research Design

This research follows a quantitative experimental research design combined with a software-prototype development approach. The main objective is to develop a district/RDHS-level dengue early-warning framework for Sri Lanka using weekly dengue case data and NASA POWER meteorological data. The proposed system is designed to predict dengue risk four weeks ahead so that possible outbreak situations can be identified before they become severe.

The analytical component focuses on constructing a longitudinal weekly dataset, examining delayed climate–dengue relationships, engineering temporal and seasonal predictors, and comparing three representative forecasting approaches: SARIMAX, LightGBM, and Long Short-Term Memory (LSTM). The software-development component focuses on converting the analytical outputs into an interactive early-warning and decision-support prototype.

The methodology maintains a common forecasting horizon and evaluation procedure across the three modelling approaches so that their performance can be compared under equivalent experimental conditions.

3.1.1 Research Approach

The research methodology consists of seven main stages:

1. **Data Acquisition:** Collect weekly dengue incidence from the Ministry of Health Weekly Epidemiological Reports and meteorological observations from the NASA POWER API for the 26 RDHS/District administrative health areas.
2. **Data Preprocessing and Integration:** Standardise epidemiological and meteorological records, convert daily climatic observations into weekly values, and align the datasets using RDHS/District area, year, and epidemiological week.
3. **Exploratory Data Analysis and Climate-Lag Analysis:** Examine temporal dengue patterns, climatic-zone variation, monsoonal behaviour, and delayed relationships between climatic variables and dengue incidence. Candidate weather lags from one to twelve weeks will be investigated.

4. **Feature Engineering:** Generate additional model inputs from the dengue and weather data. Candidate features include previous-week, two-week, and four-week dengue lags, rolling averages of dengue cases, precipitation lags, temperature lags, humidity lags, rolling weather averages, epidemiological-week and month indicators, climatic-zone information, and Southwest-monsoon, Northeast-monsoon, and inter-monsoon indicators.
5. **Comparative Modelling:** Develop SARIMAX as the statistical time-series baseline, LightGBM as the machine-learning approach, and LSTM as the deep-learning sequence model.
6. **Validation and Evaluation:** Evaluate all modelling approaches using the same four-week-ahead prediction setup and walk-forward validation procedure. Regression and outbreak-classification performance will be assessed separately.
7. **Dashboard Prototype Development:** Integrate the selected forecasting outputs into an interactive prototype that presents predicted case counts, alert status, dengue trends, and weather trends. If implementation time permits, district/RDHS risk maps and important weather-lag factors will also be presented.

3.2 Development Environment

The research environment consists of a personal-computing platform together with Python-based data-engineering and predictive-modelling technologies. The selected environment supports epidemiological-data preparation, automated meteorological-data retrieval, exploratory analysis, time-series modelling, machine learning, deep learning, validation, and dashboard development.

3.2.1 Hardware Specifications

Table 3.1 summarises the primary hardware environment used for research development and computational experiments.

Table 3.1: Hardware configuration used for research development

Component	Specification
Processor	AMD Ryzen 3 5300U
RAM	20 GB
Storage	500 GB SSD
Operating System	Windows 11

3.2.2 Software Tools

The tools and technologies used in the research were selected according to the data-processing, modelling, evaluation, explainability, and dashboard-development requirements of the study.

Python provides the main analytical environment, while Pandas and NumPy support weather-data preparation, dengue-data integration, feature generation, and analysis of dengue–weather relationships. A Python-based Streamlit application has been developed to automate communication with the NASA POWER API using the Python `requests` library.

The modelling environment includes `statsmodels` or `pmdarima` for SARIMAX-based time-series forecasting, LightGBM and scikit-learn for machine-learning development and evaluation, and TensorFlow/Keras or PyTorch for LSTM-based sequence modelling. Power BI, Matplotlib, and Plotly support dengue trends, weather patterns, lag-analysis plots, model-result visualisation, and dashboard charts. SHAP or feature-importance techniques are considered particularly for interpretation of the LightGBM model. GitHub is used for source-code management, documentation, experiment versions, and group collaboration.

Table 3.2 summarises the selected tools and their roles.

3.3 System Architecture

The proposed early-warning system is organised into three logical layers: the data layer, the business-logic and analytical layer, and the presentation layer. These layers represent the main research flow of the study: data acquisition and integration, exploratory and climate-lag analysis, feature engineering, comparative modelling, dual-stream prediction, validation, and dashboard presentation.

Figure 3.1 presents the overall architecture of the proposed framework.

3.3.1 Component 1: Data Layer

The data layer combines the two principal data streams used in the research. The epidemiological stream consists of weekly dengue case counts extracted from the Weekly Epidemiological Reports published by the Epidemiology Unit, Ministry of Health, Sri Lanka. The meteorological stream consists of precipitation, temperature, and relative-humidity observations retrieved from the NASA POWER API.

The NASA POWER queries use representative geographical coordinates associated with the 25 districts of Sri Lanka together with Kalmunai as an additional administrative reporting unit, giving 26 RDHS/District areas in total. Since the two data sources

Table 3.2: Selected tools, technologies, and frameworks

Category	Tools / Technologies	Purpose
Programming	Python	Main language for data processing, modelling, and evaluation.
Data handling	Pandas, NumPy	Preparing weather data, integrating dengue records, generating features, and analysing dengue–weather relationships.
Weather-data acquisition	NASA POWER API, Python requests	Retrieving precipitation, temperature, and relative-humidity observations using representative RDHS-area coordinates.
Statistical modelling	statsmodels / pmdarima	Developing and testing SARIMAX time-series models.
Machine learning	LightGBM, scikit-learn	Developing the LightGBM model and supporting preprocessing, validation, and metric calculation.
Deep learning	TensorFlow/Keras or PyTorch	Candidate framework for implementing the LSTM sequence model; the final framework will be confirmed during model implementation.
Analysis and visualization	Power BI, Matplotlib, Plotly	Exploratory analysis, dengue and climate trends, lag-analysis plots, model-result visualization, and dashboard charts.
Explainability	SHAP / feature importance	Interpreting influential predictors, particularly for the LightGBM model.
Dashboard development	Streamlit, React	Streamlit is used for the weather-data retrieval utility; dashboard technologies will be extended or finalized during prototype development.
Version control	GitHub	Managing source code, documentation, experiment versions, and group collaboration.

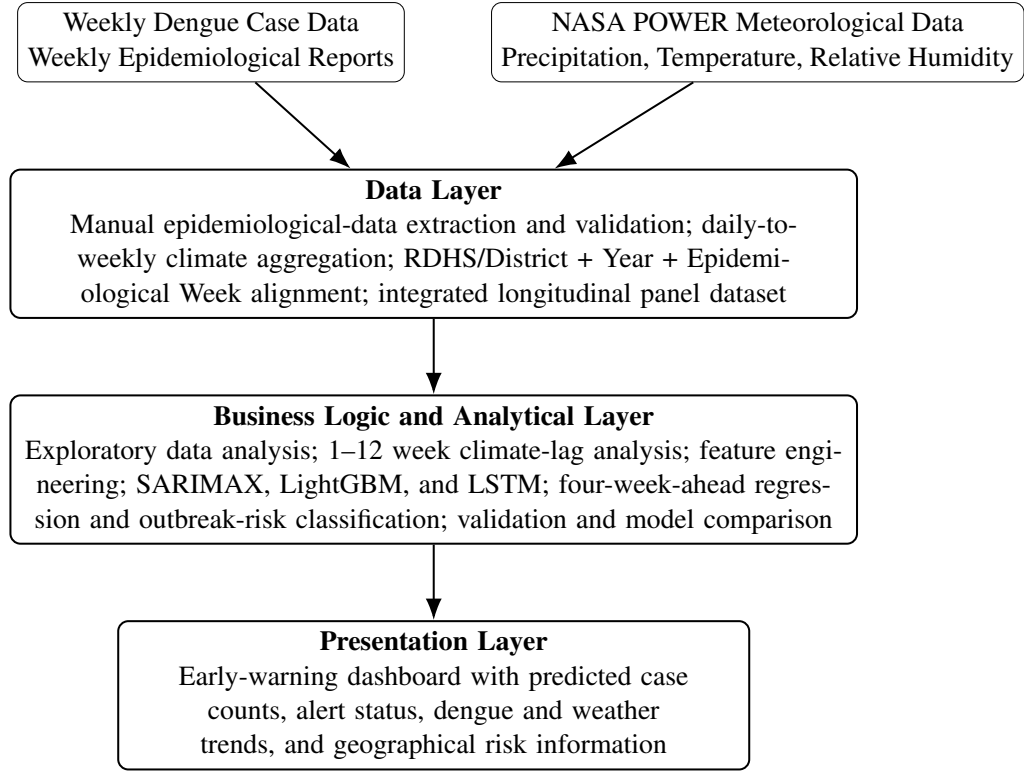


Figure 3.1: Architecture of the proposed dengue outbreak early-warning system

use different temporal resolutions and structures, both are transformed into a common RDHS/District-week representation. Daily meteorological observations are aggregated into weekly summaries and aligned with dengue incidence using RDHS/District area, year, and epidemiological week. The output of the data layer is the longitudinal panel dataset used for analysis and modelling.

3.3.2 Component 2: Business Logic Layer

The business-logic layer contains the analytical and predictive components of the system. Exploratory analysis examines dengue trends, regional variation, climatic-zone differences, seasonal behaviour, and weather patterns.

Climate-lag analysis will test precipitation, temperature, and relative-humidity values from one to twelve weeks before the corresponding dengue observations in order to investigate delayed weather–dengue relationships across RDHS/District areas and, where applicable, across the Wet, Dry, and Intermediate climatic zones.

The feature-engineering process converts the original dengue and weather data into additional model inputs. Candidate features include previous-week, two-week, and four-week dengue case values, rolling average case counts, precipitation lags, temperature lags, humidity lags, rolling weather averages, epidemiological-week and month information, Southwest-monsoon, Northeast-monsoon and inter-monsoon indicators, and climatic-zone information.

Three modelling approaches will be evaluated under the same four-week-ahead forecasting setup. SARIMAX represents the statistical time-series approach, LightGBM represents machine learning on structured lag-based features, and LSTM represents sequential deep learning from previous dengue and weather observations.

The modelling output follows a dual-stream prediction design. The regression stream predicts the expected number of dengue cases four weeks ahead. The classification stream predicts whether an RDHS/District area is likely to enter an outbreak or high-risk condition four weeks ahead. A district/RDHS-specific historical alert threshold, such as the historical mean plus two standard deviations, is considered for defining the classification target. The final threshold definition will be determined during the modelling stage.

3.3.3 Component 3: Presentation Layer

The presentation layer is designed to convert the modelling outputs into information that can be interpreted for public-health decision support. The planned dashboard will mainly present predicted dengue case counts, alert status, dengue trends, and weather trends.

If implementation time permits, district/RDHS risk maps and important contributing weather-lag factors will also be included. The dashboard remains a research decision-support prototype and is not intended to replace official epidemiological surveillance or public-health judgement.

3.4 Implementation

Implementation has been carried out incrementally according to the research workflow. At the intermediate stage, the major implemented components are epidemiological-data collection, meteorological-data automation, weekly weather aggregation, spatiotemporal data integration, and exploratory analysis. Comparative model development, model validation, outbreak classification, and complete dashboard integration form the subsequent stages.

Epidemiological-data implementation. Historical dengue records were obtained from Weekly Epidemiological Report PDF documents. Because the document layouts and table structures change across the historical period, automated extraction attempts were not sufficiently reliable. Consequently, more than 540 weekly reports were manually reviewed and the required dengue values were organised into a structured spreadsheet-based dataset indexed by year, epidemiological week, and the 26 RDHS/District administrative areas.

Meteorological-data implementation. A Python-based Streamlit application was

developed to communicate with the NASA POWER API through the Python `requests` library. The application retrieves daily precipitation, atmospheric temperature, and relative-humidity values using geographical coordinates for the 26 RDHS/District areas and allows the required date range to be selected for data retrieval.

Since NASA POWER provides daily-resolution observations, a Pandas-based processing pipeline converts these values into weekly summaries. Temperature and relative humidity are represented using weekly averages, while precipitation is represented using a cumulative weekly total. A mean-daily precipitation field is also retained in the working dataset.

Spatiotemporal data integration. The epidemiological and meteorological datasets are joined using a composite key consisting of RDHS/District area, epidemiological week, and year. This creates a longitudinal panel dataset in which each weekly observation contains the dengue incidence for an area together with the corresponding meteorological values.

Exploratory analysis and feature preparation. Power BI and Python-based analytical tools are used to inspect temporal dengue patterns, regional and climatic-zone differences, weather behaviour, seasonal fluctuations, and possible delayed weather relationships. The planned feature set includes historical dengue lags, weather lags, rolling averages, monsoon-period indicators, epidemiological week, month, and climatic zone.

Comparative modelling and decision-support implementation. The subsequent modelling stage will compare SARIMAX, LightGBM, and LSTM using the same four-week-ahead prediction objective and walk-forward validation approach. The output will contain a regression stream for future dengue case counts and a classification stream for outbreak or high-risk alerts.

The final selected model will be chosen by considering predictive performance and practical usefulness. Regression accuracy, classification performance, alert reliability, and interpretability will be considered before selecting the model for dashboard integration.

3.5 Data Collection

The study constructs a long-term longitudinal dataset covering weekly observations from January 2016 to 31 May 2026. The geographical scope contains all 26 RDHS/District administrative health areas represented in the Weekly Epidemiological Reports, corresponding to the 25 districts together with Kalmunai as an additional health-administrative reporting unit.

3.5.1 Data Sources

Epidemiological data. The primary source of dengue-incidence information is the Weekly Epidemiological Report published by the Epidemiology Unit, Ministry of Health, Sri Lanka. The reports provide weekly dengue case counts by RDHS administrative health area and are accessed through the official Weekly Epidemiological Report repository:

```
https://www.epid.gov.lk/weekly-epidemiological-report/  
weekly-epidemiological-report
```

For each reporting week, the extracted epidemiological information is indexed using year, epidemiological week, and RDHS/District area.

Meteorological data. Precipitation, atmospheric temperature, and relative-humidity observations are obtained through the NASA POWER API. Representative geographical coordinates are used for the 26 RDHS/District areas to retrieve daily meteorological observations, which are subsequently transformed to weekly values before integration with the dengue-surveillance dataset.

The research initially considered obtaining historical meteorological observations from the Sri Lanka Department of Meteorology. However, the required continuous localised data across the complete study period and geographical coverage could not be obtained within the available undergraduate research budget. NASA POWER was therefore selected as the alternative meteorological source used in this research.

Human-mobility information was also considered in the initial research design. The team explored access to telecommunications-based mobility information, but the required data could not be obtained because of privacy, security, and institutional data-governance restrictions. Human mobility was therefore removed from the implemented predictor set, and the research proceeded using dengue incidence and meteorological variables.

3.5.2 Data Preprocessing

Data preprocessing is performed separately for the epidemiological and meteorological sources before integration.

Epidemiological-data extraction and standardisation. The historical Weekly Epidemiological Reports contain structural and formatting differences across years. Automated extraction attempts produced inconsistent results for some reports. To preserve the structure of the historical dataset, the required dengue tables were manually reviewed and transferred into a standardised spreadsheet-based structure. RDHS/District names, year values, and epidemiological-week identifiers were standardised before integration.

Meteorological aggregation. NASA POWER observations are retrieved at daily resolution and grouped into seven-day epidemiological calendar windows. The aggregation procedure calculates:

- weekly mean atmospheric temperature;
- weekly mean relative humidity; and
- total weekly precipitation as the sum of the seven daily precipitation values.

A mean-daily precipitation value is also retained as a separate field in the working dataset.

Spatiotemporal alignment. A composite key based on RDHS/District area, year, and epidemiological week is used to merge dengue and meteorological records. This ensures that the dengue incidence for a particular reporting week is associated with the meteorological observations assigned to the same geographical area and epidemiological week.

Missing-value handling. Missing dengue values identified during the manual construction of the historical dataset are not automatically treated as zero. Linear interpolation is applied within the corresponding RDHS/District series. A missing dengue value is estimated using the closest preceding and succeeding valid observations so that chronological continuity is retained in the panel dataset.

Dataset structure. The resulting dataset is organised as a longitudinal RDHS/District-week panel. For each epidemiological week, the dataset represents the 26 administrative areas as separate records. The working data dictionary contains the temporal, geographical, meteorological, and epidemiological variables required for the subsequent analysis and modelling.

Table 3.3 summarises the principal variables.

Table 3.3: Data dictionary for the integrated RDHS-week dataset

Variable	Data Type	Description
Year	Integer	Calendar year of the observation, covering 2016–2026.
Month	Text	Calendar month associated with the surveillance week.
Week No	Integer	Continuous week index across the longitudinal study period.
Epi Week	Integer	Epidemiological week number used to align surveillance and climatic records.

Table 3.3 – continued from previous page

Variable	Data Type	Description
Week Start	Date	Starting date of the corresponding surveillance week.
Week End	Date	Ending date of the corresponding surveillance week.
District / RDHS Area	Text	Administrative health area represented in the Weekly Epidemiological Reports.
Zone	Text	Climatic-zone classification: Wet, Dry, or Intermediate.
Temperature	Float	Weekly mean atmospheric temperature derived from NASA POWER daily observations.
Relative Humidity	Float	Weekly mean relative-humidity value derived from NASA POWER daily observations.
Mean Daily Precipitation	Float	Mean daily precipitation calculated across the corresponding seven-day period.
Total Weekly Precipitation	Float	Sum of daily precipitation values across the corresponding seven-day period.
Dengue Cases	Integer	Total reported dengue cases for the RDHS area and epidemiological week.

3.6 Testing and Evaluation Strategy

The study uses walk-forward validation for model evaluation. Since dengue forecasting is treated as a time-series problem, random train-test splitting is not used because it can mix past and future observations. Instead, the study uses walk-forward validation in which the models are trained on past data and tested on later data.

All three modelling approaches will be evaluated using the same dataset, four-week-ahead prediction horizon, and validation method. This provides a common basis for comparing SARIMAX, LightGBM, and LSTM.

3.6.1 Evaluation Metrics

The dual-stream prediction design is evaluated using separate regression and classification metrics.

For the regression stream, MAE, RMSE, MAPE, and forecast bias will be used to evaluate the four-week-ahead dengue case-count predictions.

Table 3.4: Regression evaluation metrics

Metric	Purpose
Mean Absolute Error (MAE)	Measures the average absolute difference between predicted and observed dengue case counts.
Root Mean Squared Error (RMSE)	Penalizes larger forecasting errors more strongly than MAE.
Mean Absolute Percentage Error (MAPE)	Expresses forecasting error as a percentage of the observed value.
Forecast Bias	Indicates whether the model tends to systematically overpredict or underpredict dengue cases.

For the classification stream, Precision, Recall, F1-score, and ROC-AUC will be used to evaluate outbreak or high-risk alerts.

Table 3.5: Classification evaluation metrics

Metric	Purpose
Precision	Measures the proportion of predicted outbreak alerts that correspond to actual high-risk situations.
Recall	Measures the proportion of actual outbreak or high-risk situations successfully detected by the model.
F1-score	Provides a balanced measure combining precision and recall.
ROC-AUC	Measures the model's ability to distinguish between alert and non-alert situations across classification thresholds.

Recall is considered particularly important for outbreak prediction because missing a real outbreak can be more harmful than issuing a false warning. Final model selection will therefore consider both numerical prediction performance and the ability to detect high-risk dengue situations.

3.6.2 Testing Approach

The primary testing approach is walk-forward validation. In this procedure, a model is trained using historical observations and evaluated on a later period. The training window is then expanded forward and the process is repeated.

The illustrative validation setup is:

Training Period	Testing Period
2016–2020	2021
2016–2021	2022
2016–2022	2023
2016–2023	2024

Before validation, the dataset is checked to ensure that dengue observations are correctly aligned with the same RDHS/District area and epidemiological week as the meteorological variables.

The same dataset, prediction horizon, and validation method will be applied to the three candidate modelling approaches. The final selected model will be chosen based on regression accuracy, classification performance, outbreak-alert reliability, interpretability, and practical usefulness for the proposed dashboard. A model with slightly lower numerical accuracy may still be useful if it provides better outbreak detection and interpretability for an early-warning application.

3.7 Chapter Summary

This chapter described the methodology used for the proposed district/RDHS-level dengue outbreak early-warning framework. The research combines a quantitative experimental design with software-prototype development and uses weekly dengue surveillance records together with NASA POWER precipitation, temperature, and relative-humidity observations.

The methodology follows seven main stages: data acquisition, preprocessing and integration, exploratory analysis and one-to-twelve-week climate-lag analysis, feature engineering, comparative modelling using SARIMAX, LightGBM, and LSTM, walk-forward validation and evaluation, and dashboard prototype development.

The modelling framework follows a dual-stream prediction design. It is intended to produce four-week-ahead dengue case-count forecasts through regression and outbreak or high-risk alerts through classification. Regression and classification results will be evaluated separately before the most suitable model is selected for integration with the decision-support dashboard.

Chapter 4

Results and Analysis

4.1 Overview of Results

This chapter presents the results available at the intermediate stage of the research. At this stage, the main completed analytical outputs consist of the integrated RDHS-week longitudinal dataset and the exploratory analysis performed using dengue incidence and NASA POWER meteorological variables.

The analysis covers weekly observations from January 2016 to 31 May 2026 across all 26 RDHS administrative health areas. The preliminary results focus on national dengue time-series behaviour, climatic-zone differences, national weekly precipitation patterns, and the weekly behaviour of temperature and relative humidity.

For the national climate visualisations, the meteorological values are first prepared at RDHS-week level. The national precipitation series represents the average of the RDHS-level weekly precipitation totals for each week, while the temperature and relative-humidity series represent the corresponding averages across the 26 RDHS areas. National dengue incidence is represented as the sum of the weekly dengue cases across the same RDHS areas.

These national aggregates are used only for exploratory visualisation. The underlying modelling dataset retains the individual RDHS-week observations so that local climatic relationships and time lags can be analysed separately.

Full comparative evaluation of SARIMAX, LightGBM, and LSTM had not yet been completed at the time of the intermediate analysis. Therefore, this chapter does not present incomplete or unverified final model-performance values. Instead, it documents the preliminary evidence generated from the completed data-engineering and exploratory-analysis stages and identifies the results that remain to be obtained during the modelling phase.

4.2 Performance Evaluation

The intermediate research stage established the evaluation framework for SARIMAX, LightGBM, and LSTM but did not yet produce a complete and reproducible set of final four-week-ahead model-performance results.

The exploratory analysis nevertheless demonstrates substantial temporal and spatial variation in the dataset. The dengue time series contains both lower-incidence periods and sharp epidemic peaks, while climatic-zone differences and the distinct temporal behaviour of the meteorological variables suggest that a purely static or nationally aggregated approach may not adequately represent local dengue dynamics.

The final comparative modelling stage will evaluate all three models using the time-aware walk-forward validation procedure defined in Chapter 3. Regression performance will be measured using MAE, RMSE, MAPE, and forecast bias, while outbreak classification will be evaluated using Precision, Recall, F1-score, and ROC-AUC.

No final numerical ranking between SARIMAX, LightGBM, and LSTM is reported at this stage because the required comparative training and validation experiments had not yet been completed.

4.3 Functional Testing Results

Formal end-to-end functional testing of the final early-warning dashboard had not yet been completed at the intermediate stage. Functional verification was therefore limited primarily to the implemented data-acquisition and preprocessing pipeline.

The completed implementation demonstrated that:

- historical weekly dengue observations could be structured consistently across the 26 RDHS administrative health areas;
- the NASA POWER API retrieval utility could obtain daily precipitation, temperature, and relative-humidity observations for the selected geographical coordinates;
- daily meteorological observations could be converted into compatible weekly summaries;
- dengue and meteorological records could be aligned using RDHS area, year, and epidemiological week; and
- the resulting integrated dataset could support exploratory visualisation and subsequent feature-engineering activities.

These checks confirm the functional readiness of the data pipeline for the subsequent modelling stage. They should not, however, be interpreted as formal software-system testing of the final dashboard, which remains a later implementation activity.

4.4 User Study Results

No formal user study was conducted during the intermediate stage of the research. The final decision-support dashboard had not yet reached a sufficiently complete state for meaningful usability evaluation by public-health users.

Consequently, no participant counts, questionnaire scores, usability ratings, or user-satisfaction results are reported in this chapter. Any future user evaluation will be documented only after the dashboard prototype and its prediction outputs are sufficiently complete for testing.

4.5 Comparison with Existing Solutions

A direct numerical comparison with existing dengue forecasting systems is not yet appropriate because the final SARIMAX, LightGBM, and LSTM experiments have not been completed under the common validation framework.

The intermediate results instead establish the conditions required for a later fair comparison. The developed dataset provides common weekly inputs across all 26 RDHS areas, and all three candidate models are intended to use the same four-week-ahead forecasting objective and time-aware evaluation procedure.

This approach differs from comparing performance values reported independently by previous studies, since those studies often use different geographical areas, temporal resolutions, predictor sets, and validation procedures. The final comparison in this research will therefore focus primarily on the relative performance of SARIMAX, LightGBM, and LSTM when evaluated under the same Sri Lankan RDHS-level experimental environment.

4.6 Statistical Analysis

At the intermediate stage, the statistical analysis is primarily descriptive and exploratory. No inferential test such as a t-test is reported because the completed analysis does not provide evidence for such a comparison.

National dengue time-series behaviour. Figure 4.1 presents the aggregated dengue case trend across the continuous study timeline. The series demonstrates highly irregular epidemic behaviour rather than a constant level, with substantial variation and several pronounced epidemic peaks. A major surge is visible around continuous week numbers 70–90, where aggregated weekly cases exceed approximately 10,000. A second major period is visible near week number 200, where case counts approach approximately 5,000.



Figure 4.1: Aggregated dengue cases across the continuous weekly study timeline

These sharp changes indicate that the forecasting problem must account for both longer-term temporal structure and sudden epidemic peaks. However, the figure alone does not establish that one specific model family will necessarily outperform another; this must be determined through comparative validation.

Climatic-zone distribution. The cumulative dengue burden also differs substantially between the Wet, Dry, and Intermediate climatic zones. Figure 4.2 shows the zone-level distribution obtained from the exploratory analysis.

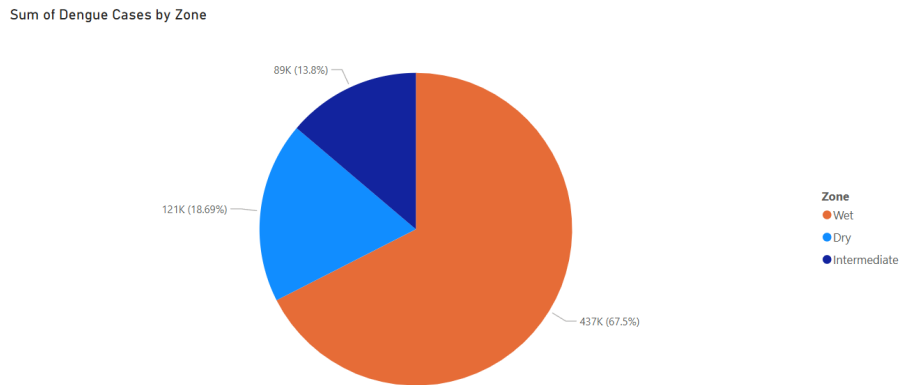


Figure 4.2: Distribution of cumulative dengue cases by climatic zone

The approximate cumulative dengue-case distribution across the three climatic zones is summarised below.

The Wet Zone accounts for approximately two-thirds of the cumulative dengue cases represented in the exploratory dataset. The Dry and Intermediate zones contribute substantially smaller shares. This difference provides descriptive motivation for retaining climatic-zone information in later analysis and feature engineering, although the aggregated proportions alone do not establish a causal climatic effect.

National weekly precipitation and dengue behaviour. Figure 4.3 compares the

Table 4.1: Approximate cumulative dengue-case distribution by climatic zone

Climatic Zone	Approximate Cases	Share (%)
Wet Zone	437,000	67.50
Dry Zone	121,000	18.69
Intermediate Zone	89,000	13.80

national weekly average precipitation with the national weekly dengue-case total.

For each epidemiological week, precipitation is first calculated as a weekly total for each individual RDHS area. The values plotted in Figure 4.3 then represent the average of those RDHS-level weekly totals across the 26 RDHS areas. Dengue cases, in contrast, are summed across the 26 areas to represent the national weekly dengue burden.

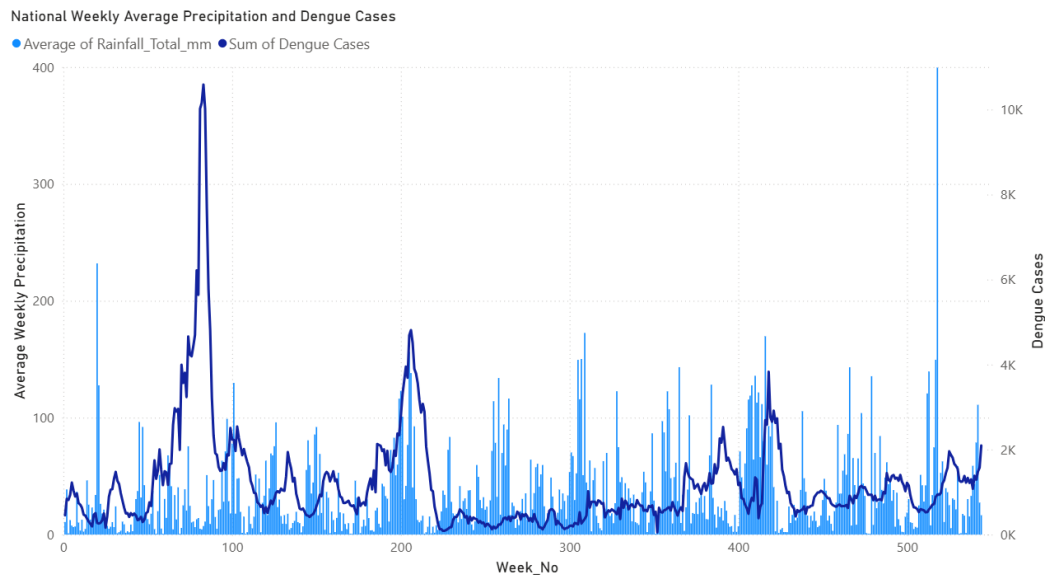


Figure 4.3: National weekly average precipitation and dengue cases across the study timeline

The precipitation series exhibits considerably greater week-to-week variability than the national temperature and relative-humidity series, with several distinct high-precipitation periods visible across the study timeline. The dengue series also contains major epidemic peaks, but these peaks do not consistently coincide with precipitation peaks during the same week.

Some periods visually suggest that changes in precipitation may precede later changes in dengue incidence. However, the nationally averaged figure does not provide sufficient evidence to estimate a precise climate-to-dengue lag. Averaging precipitation across 26 RDHS areas can mask geographically different rainfall patterns and local dengue responses.

Therefore, Figure 4.3 is interpreted only as exploratory evidence that same-week

precipitation alone is unlikely to explain the observed dengue pattern. Formal lag analysis will be conducted at RDHS level using candidate lags from one to twelve weeks as defined in Chapter 3.

National weekly temperature and humidity behaviour. Figures 4.4 and 4.5 compare the national weekly averages of temperature and relative humidity with the national weekly dengue-case total.

For each week, the temperature value represents the average of the corresponding RDHS-level weekly mean temperatures across the 26 RDHS areas. Relative humidity is summarised in the same manner using the RDHS-level weekly mean humidity values.

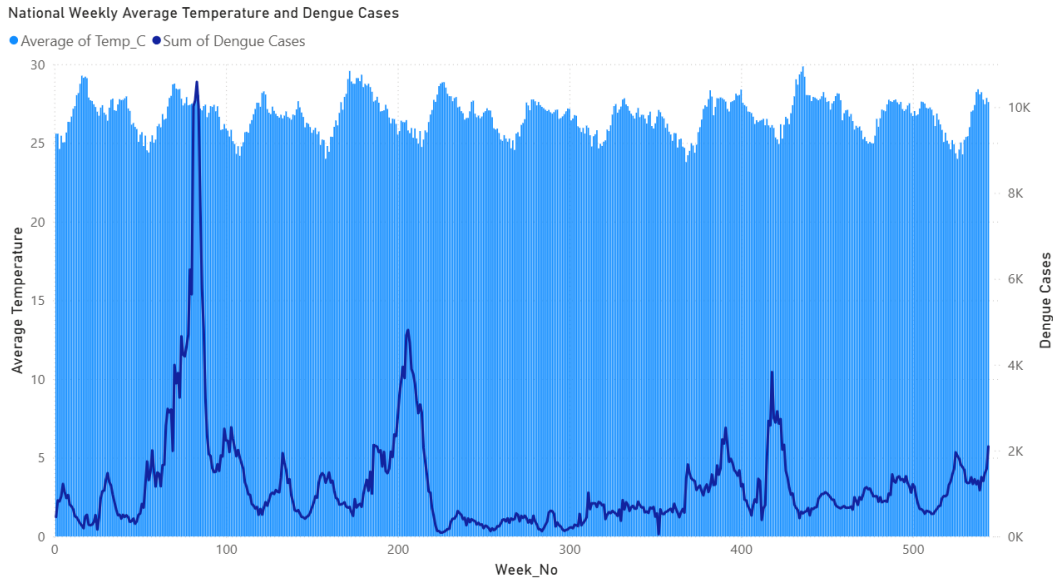


Figure 4.4: National weekly average temperature and dengue cases across the study timeline

Compared with precipitation and dengue incidence, the nationally averaged temperature and relative-humidity profiles vary within narrower ranges throughout the study timeline. The visualisations do not show an obvious same-week one-to-one relationship between changes in these national climate averages and the major dengue peaks.

This observation should not be interpreted as evidence that temperature or humidity are unimportant for dengue transmission. National averaging can smooth substantial differences between RDHS areas, while climatic effects may also be delayed, seasonal, nonlinear, or interactive with other variables.

Consequently, the predictive contribution of temperature and relative humidity will be assessed using the original RDHS-level weekly observations, lagged variables, seasonal indicators, and multivariable forecasting models rather than relying on the national aggregate visualisations alone.

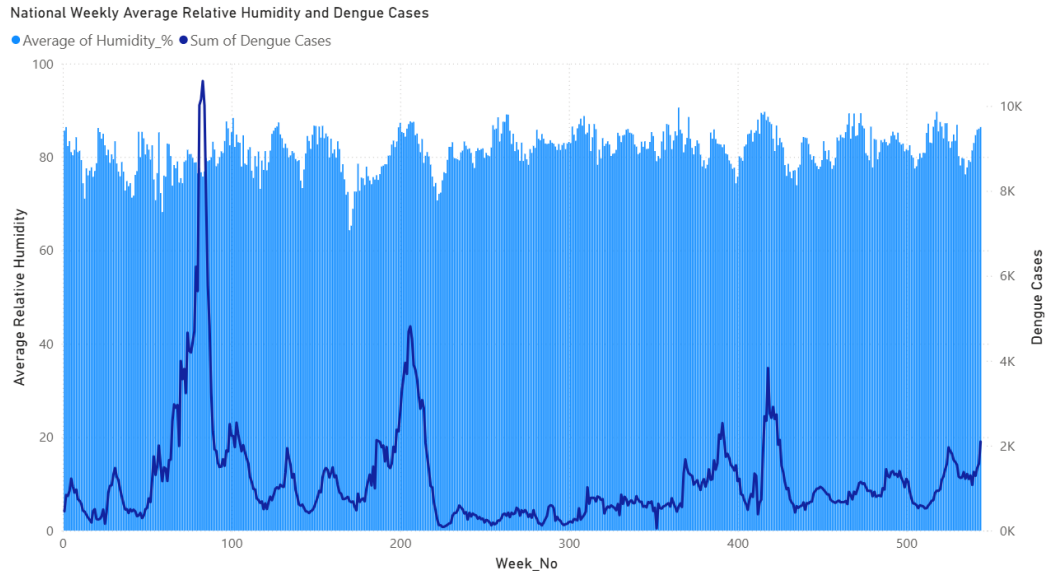


Figure 4.5: National weekly average relative humidity and dengue cases across the study timeline

4.7 Discussion

4.7.1 Key Findings

The intermediate analysis produces four main findings:

1. **Dengue incidence is strongly non-uniform over time.** The national weekly series contains substantial epidemic peaks separated by lower-incidence periods, confirming that the prediction task must accommodate strong temporal variability.
2. **Dengue burden is spatially unequal across climatic zones.** The Wet Zone contributes approximately 67.5% of the cumulative cases represented in the exploratory dataset, compared with approximately 18.69% in the Dry Zone and 13.8% in the Intermediate Zone.
3. **Precipitation and dengue do not display a simple same-week relationship.** National weekly average precipitation varies considerably across the study timeline, while major dengue peaks do not consistently occur during the same weeks as precipitation peaks. This supports further investigation of delayed climate effects, but a precise lag cannot be inferred from the nationally aggregated chart alone.
4. **National average temperature and humidity are comparatively stable.** Their aggregate weekly profiles vary within narrower ranges than precipitation and dengue incidence. No clear same-week one-to-one relationship with the major

dengue peaks is evident from the national visualisations. RDHS-level lagged and multivariable analysis is therefore required.

These findings support the overall structure of the proposed comparative modelling framework but do not yet determine which of SARIMAX, LightGBM, or LSTM will provide the strongest forecasting performance.

4.7.2 Addressing Research Questions

The status of the research questions at the intermediate stage is summarised below.

RQ1: Climate relationships and RDHS-specific time lags.

RQ1 is **partially addressed**. The exploratory analysis demonstrates that the weekly climatic variables and dengue incidence have different temporal patterns and that a simple same-week relationship is not evident from the national aggregate visualisations. The climatic-zone analysis also demonstrates substantial geographical differences in cumulative dengue burden.

However, Figures 4.3, 4.4, and 4.5 summarize climatic variables across all 26 RDHS areas and therefore cannot establish local optimal time lags. The full RDHS-specific one-to-twelve-week lag analysis has not yet been completed. Consequently, no final RDHS-specific climate-to-dengue lag values are claimed at this stage.

RQ2: Comparison of SARIMAX, LightGBM, and LSTM.

RQ2 is **not yet fully answered**. The comparative modelling environment has been designed, but final walk-forward validation results for all three candidate models are not yet available.

RQ3: Four-week-ahead outbreak-risk classification.

RQ3 is **not yet fully answered**. The dual regression/classification design and candidate alert-threshold approach have been defined, but the outbreak-classification models have not yet been fully trained and evaluated.

RQ4: Dashboard-based decision support.

RQ4 is **partially addressed at the design level**. The intended dashboard outputs have been specified, and a Streamlit-based meteorological-data utility has already been implemented. However, the complete early-warning dashboard containing final prediction and alert outputs has not yet been evaluated with intended users.

4.7.3 Limitations of Results

The preliminary results should be interpreted with the following limitations:

- The reported findings are primarily exploratory and descriptive rather than final predictive-model results.

- Figures 4.3, 4.4, and 4.5 are national aggregate visualisations. Meteorological values are averaged across the 26 RDHS areas, whereas dengue cases are summed to represent the national weekly burden. These summaries may hide important local climatic and epidemiological differences.
- The national precipitation visualisation may suggest temporal separation between precipitation events and dengue peaks, but it does not provide sufficient evidence for a precise climate-to-dengue lag. Formal lag values must be determined from RDHS-specific analysis.
- Climatic-zone case proportions describe the distribution represented in the historical dataset but do not independently establish causation between climatic zone and dengue incidence.
- Missing dengue observations were handled through within-RDHS interpolation during dataset preparation. The influence of imputed observations on model evaluation must be considered during final validation.
- Human mobility, detailed entomological indicators, socioeconomic conditions, and local vector-control activities are not included in the current dataset.
- Final SARIMAX, LightGBM, and LSTM performance values and outbreak-classification metrics are not yet available at the intermediate stage.
- The complete decision-support dashboard has not yet undergone a formal user study.

4.8 Chapter Summary

This chapter presented the results available from the completed data-engineering and exploratory-analysis stages of the research. The national weekly dengue series demonstrates substantial epidemic variability, including major historical surges, while the climatic-zone analysis shows a strong concentration of cumulative dengue cases within the Wet Zone.

The national climate visualisations were constructed using weekly RDHS-level climatic summaries. National precipitation represents the average of the weekly precipitation totals across the 26 RDHS areas, while temperature and relative humidity represent national averages of the corresponding RDHS-level weekly means. Weekly dengue cases are summed across the RDHS areas to represent national disease burden.

The precipitation series shows substantial temporal variability, but major precipitation peaks and dengue peaks do not consistently occur during the same weeks. This indicates that same-week precipitation alone is insufficient to describe dengue behaviour and supports the planned investigation of delayed climatic effects. However, the na-

tional aggregate chart cannot establish a precise climate-to-dengue lag.

National weekly average temperature and relative humidity display comparatively narrower variation and do not show an obvious same-week relationship with the major dengue peaks. Since national aggregation can mask local climatic differences, the original RDHS-level observations will be retained for climate-lag analysis and predictive modelling.

These findings provide descriptive motivation for retaining climatic-zone information and for investigating delayed climatic relationships using the original RDHS-level weekly observations. However, they do not yet establish the most informative climate lags or the best-performing forecasting model. Final conclusions regarding SARI-MAX, LightGBM, LSTM, four-week-ahead case forecasting, outbreak classification, and dashboard effectiveness require completion of the comparative walk-forward validation and system-evaluation stages.

Chapter 5

Conclusion

5.1 Summary of Research

This research investigates the development of a district-level dengue outbreak early-warning system for Sri Lanka using historical dengue incidence and climatic information. The study is motivated by the need to move beyond retrospective dengue surveillance toward a data-driven framework capable of supporting localised and proactive public-health decision making.

The research uses a longitudinal weekly dataset covering January 2016 to 31 May 2026 across all 26 Regional Director of Health Services (RDHS) administrative health areas represented in the national Weekly Epidemiological Reports. Historical weekly dengue case counts are combined with precipitation, temperature, and relative-humidity observations obtained through the NASA POWER API.

At the current stage of the research, the major completed activities include historical dengue-data preparation, meteorological-data acquisition and weekly aggregation, spatiotemporal integration of the two data sources, development of a consistent RDHS-week dataset, and exploratory analysis of temporal, geographical, and climatic patterns.

The research framework is designed to extend these completed stages through RDHS-specific climate-lag analysis, feature engineering, and comparative evaluation of three forecasting approaches: SARIMAX, LightGBM, and Long Short-Term Memory (LSTM). The intended predictive output consists of both four-week-ahead dengue case-count forecasts and localised outbreak-risk alerts.

The selected predictive outputs are ultimately intended to be integrated into an interactive decision-support dashboard that presents forecasts, risk status, temporal trends, climatic information, and geographical differences between RDHS areas.

5.2 Key Findings and Contributions

The results available at the intermediate stage provide several findings relevant to the continuing development of the proposed early-warning framework.

Temporal variation in dengue incidence. The national weekly dengue time series shows substantial variation over the study period, including several major epidemic

peaks separated by lower-incidence periods. This indicates that dengue forecasting must account for changing temporal patterns rather than assuming a constant level of disease activity.

Geographical variation across climatic zones. The exploratory analysis shows a substantial difference in cumulative dengue burden between the Wet, Dry, and Intermediate climatic zones. Approximately 67.5% of the cumulative cases represented in the exploratory dataset occur within the Wet Zone, compared with approximately 18.69% in the Dry Zone and 13.8% in the Intermediate Zone. These aggregated proportions do not establish causality, but they provide descriptive motivation for retaining geographical and climatic-zone information in subsequent analysis.

Different temporal behaviour of climatic variables. National weekly average precipitation displays considerable week-to-week variability, while national average temperature and relative humidity remain within comparatively narrower ranges. The national visualisations do not demonstrate a simple same-week relationship between the climatic variables and major dengue peaks. This supports further investigation of delayed climatic effects and local RDHS-specific relationships rather than relying solely on simultaneous national averages.

Need for localised climate-lag analysis. The exploratory results show that nationally aggregated climatic visualisations cannot determine a reliable local climate-to-dengue lag. The research therefore retains the original RDHS-week observations and will investigate candidate precipitation, temperature, and humidity lags from one to twelve weeks separately at local level.

Based on the work completed so far, the main contributions of the research are:

1. **A unified RDHS-level longitudinal dataset.** Historical weekly dengue surveillance data and NASA POWER meteorological observations have been transformed into a common weekly spatiotemporal structure covering all 26 RDHS administrative health areas.
2. **A reproducible meteorological-data acquisition workflow.** A Python-based Streamlit utility was developed to retrieve daily precipitation, temperature, and relative-humidity observations from the NASA POWER API for representative RDHS-area coordinates and prepare them for weekly aggregation.
3. **National and regional exploratory analysis.** The completed analysis identifies major temporal dengue patterns, substantial differences between climatic zones, and distinct temporal characteristics of precipitation, temperature, and relative humidity.
4. **A common comparative-modelling framework.** The methodology defines a consistent experimental environment for comparing SARIMAX, LightGBM, and

LSTM using the same weekly dataset, four-week-ahead forecasting objective, and time-aware walk-forward validation procedure.

5. **A dual-stream early-warning design.** The proposed framework combines numerical dengue case forecasting with localised outbreak-risk classification rather than relying on a single predictive output.
6. **A pathway from prediction to decision support.** The system architecture connects epidemiological and climatic data preparation with predictive modelling and a planned dashboard layer intended to translate model outputs into understandable public-health information.

These contributions describe the completed foundation and designed framework of the research. They should not be interpreted as evidence that a final best-performing forecasting model or fully validated early-warning system has already been established.

5.3 Limitations

Several limitations affect the interpretation and generalisation of the current research.

- **Meteorological-data source:** Continuous official historical weather observations at the required geographical and temporal coverage could not be obtained from the Sri Lanka Department of Meteorology within the available research resources. NASA POWER is therefore used as a consistent alternative climatic source. Although suitable for large-scale longitudinal analysis, it may not represent every local micro-climatic condition observable through ground-based weather stations.
- **Historical surveillance-data extraction:** Weekly Epidemiological Reports are published as PDF documents and their structures vary across the historical period. A substantial amount of dengue data therefore required manual extraction and verification. Although validation procedures were used, manual processing remains potentially susceptible to transcription errors.
- **Missing dengue observations:** Missing values identified during historical dataset construction were handled using within-RDHS linear interpolation to maintain chronological continuity. Imputed observations are not equivalent to directly reported case counts and their influence must be considered during final model training and evaluation.
- **Limited predictor coverage:** The present dataset primarily represents historical dengue incidence and climatic variables. Potential determinants such as human mobility, detailed entomological indices, socioeconomic conditions, waste-

management conditions, local vector-control interventions, and population-movement patterns are not comprehensively represented.

- **Human-mobility data exclusion:** Telecommunications-based mobility information was considered in the initial research design but could not be obtained because of privacy, security, and institutional data-governance restrictions.
- **National aggregate exploratory visualisations:** The national climate figures average meteorological values across all 26 RDHS areas while dengue cases are summed to represent national disease burden. These visualisations are useful for exploratory overview but may hide important local climatic and epidemiological variation.
- **Incomplete predictive evaluation at the intermediate stage:** The complete comparative walk-forward evaluation of SARIMAX, LightGBM, and LSTM has not yet been completed. Consequently, the research does not currently claim that any one of these models is superior.
- **Incomplete outbreak-classification evaluation:** The classification-based alert mechanism and RDHS-specific outbreak threshold require further implementation and validation before reliable alert-performance conclusions can be reported.
- **Prototype status of the dashboard:** The complete decision-support dashboard has not yet undergone formal usability or operational evaluation with intended public-health users. Therefore, no claim is made regarding its final usability or effectiveness in real-world health-service operations.

5.4 Future Work

Future work is divided into short-term activities required to complete the current research and longer-term extensions that could improve the system beyond the present undergraduate research scope.

5.4.1 Short-term Improvements

The immediate continuation of the research should focus on completing and validating the analytical and system components already defined in the methodology.

1. **Complete RDHS-specific climate-lag analysis.** Precipitation, temperature, and relative humidity should be evaluated using candidate lags from one to twelve weeks for each RDHS area. This will determine whether the strongest climate-to-dengue relationships differ geographically and between climatic variables.

2. **Complete feature engineering.** The modelling dataset should incorporate historical dengue lags, climatic lags, rolling statistics, epidemiological-week information, month, climatic zone, and relevant monsoon-related variables while ensuring that future information is not introduced into model inputs.
3. **Implement and optimise the three forecasting approaches.** SARIMAX, LightGBM, and LSTM should be developed using the common four-week-ahead forecasting objective defined in Chapter 3.
4. **Conduct walk-forward comparative validation.** The three modelling approaches should be evaluated under the same time-aware validation procedure using MAE, RMSE, MAPE, and forecast bias for case-count prediction.
5. **Finalise outbreak-risk classification.** The RDHS-specific outbreak or high-risk threshold should be formally defined and validated. Classification performance should then be evaluated using Precision, Recall, F1-score, and ROC-AUC.
6. **Select the most suitable predictive approach.** Final model selection should consider forecasting accuracy, outbreak detection, alert reliability, interpretability, and practical suitability for integration with the decision-support system.
7. **Complete dashboard integration.** The selected forecasting and classification outputs should be integrated with the dashboard to present four-week-ahead predicted cases, risk status, historical trends, climatic information, and geographical risk patterns.
8. **Perform system and usability evaluation.** After the dashboard is sufficiently complete, functional testing and an appropriate user evaluation should be conducted before claims regarding practical usability are made.

5.4.2 Long-term Research Directions

Beyond the current project scope, several extensions could strengthen the forecasting and operational value of the proposed framework.

- **Integration of additional epidemiological predictors:** Future studies could investigate entomological indices, mosquito-density information, vector-control activities, and other disease-surveillance indicators where reliable data become available.
- **Human-mobility information:** If appropriately anonymized and institutionally approved mobility data become accessible, population-movement patterns could be investigated as additional predictors of spatial dengue transmission.
- **Ground-based meteorological observations:** NASA POWER data could be

compared or combined with official weather-station observations to investigate whether finer local climatic measurements improve RDHS-level forecasting.

- **Socioeconomic and environmental factors:** Future work could include population density, urbanization, sanitation conditions, land use, waste-management information, and other spatial determinants that may contribute to dengue risk.
- **Prospective operational validation:** After retrospective model development, the system should be evaluated prospectively using newly arriving surveillance data to determine whether four-week-ahead forecasts remain reliable under real operational conditions.
- **Integration with public-health information systems:** With appropriate institutional approval, future versions could investigate automated data updates and integration with official surveillance and health-service workflows.

5.5 Recommendations

Based on the findings and limitations of the current research, several recommendations are made for the continuation and potential application of the proposed early-warning framework.

- RDHS-level information should be retained throughout predictive analysis rather than replacing local observations with national climatic averages, because national aggregation can mask important regional differences.
- Climate-to-dengue relationships should be validated using time-lagged local analysis rather than assuming that precipitation, temperature, or humidity have an immediate same-week effect.
- All forecasting models should be evaluated using time-aware validation to prevent information from future observations from influencing predictions of earlier periods.
- Model comparison should use the same dataset, forecasting horizon, and evaluation procedure so that differences in performance reflect the modelling approaches rather than differences in experimental setup.
- Outbreak alerts should not be selected solely on the basis of overall classification accuracy. Particular attention should be given to Recall because failure to identify a true high-risk period may be more consequential than issuing an unnecessary warning.
- The influence of missing and interpolated epidemiological observations should be considered during final model training and evaluation.

- Before operational public-health use, prediction accuracy, alert thresholds, dashboard usability, and system reliability should be validated beyond the current research-prototype environment.

5.6 Concluding Remarks

The work completed so far demonstrates the feasibility of constructing a national-scale weekly RDHS-level research dataset by integrating Sri Lankan dengue-surveillance records with consistently available climatic observations. The completed data-engineering and exploratory-analysis stages reveal substantial temporal and geographical variation in dengue incidence and demonstrate why local climatic relationships should be investigated rather than assuming a single national pattern.

The study has also established a structured methodology for evaluating delayed climatic effects and comparing statistical, machine-learning, and deep-learning forecasting approaches under a common four-week-ahead framework. However, the current evidence does not yet justify claims regarding the best forecasting model, final outbreak-detection performance, or operational effectiveness of the dashboard.

Completion of the RDHS-specific lag analysis, SARIMAX–LightGBM–LSTM comparison, outbreak-classification evaluation, and dashboard validation is therefore required before final conclusions regarding predictive performance can be made.

Nevertheless, the integrated dataset, data-acquisition workflow, exploratory findings, comparative modelling design, and decision-support architecture provide a practical foundation for the remaining research and for future development of localised dengue early-warning approaches in Sri Lanka.

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Appendices

Appendix A

Data Preparation Reference

This appendix provides supplementary details regarding the construction of the integrated RDHS-week dataset used in the research. The information presented here supports the data-collection and preprocessing methodology described in Chapter 3.

A.1 Data Sources and Study Coverage

The research integrates two principal data sources.

Epidemiological data. Historical dengue incidence is obtained from the Weekly Epidemiological Reports published by the Epidemiology Unit, Ministry of Health, Sri Lanka. The reports provide weekly dengue case counts by Regional Director of Health Services (RDHS) administrative health area.

The official repository used for data collection is:

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https://www.epid.gov.lk/weekly-epidemiological-report/  
weekly-epidemiological-report
```

The study period extends from January 2016 to 31 May 2026 and covers all 26 RDHS/District administrative reporting areas represented in the reports, corresponding to the 25 districts together with Kalmunai as an additional health-administrative reporting unit.

Meteorological data. Precipitation, atmospheric temperature, and relative humidity are obtained from the NASA POWER API using representative geographical coordinates associated with the RDHS areas. NASA POWER provides daily meteorological observations, which are converted into weekly summaries before integration with the dengue surveillance records.

A.2 Weekly Meteorological Aggregation

The epidemiological data are reported weekly, while NASA POWER observations are retrieved at daily resolution. Daily meteorological observations are therefore transformed to a compatible weekly resolution.

Table A.1 summarises the transformation rules.

Table A.1: Weekly aggregation rules for the principal research variables

Variable	Source Resolution	Weekly Representation
Dengue cases	Weekly	Reported weekly dengue-case count for the corresponding RDHS area and epidemiological week.
Precipitation	Daily	Daily precipitation values are summed across the corresponding seven-day period to obtain total weekly precipitation.
Temperature	Daily	Daily atmospheric-temperature values are averaged to obtain weekly mean temperature.
Relative humidity	Daily	Daily relative-humidity values are averaged to obtain weekly mean relative humidity.

A mean-daily precipitation value is also retained as a separate field in the working dataset.

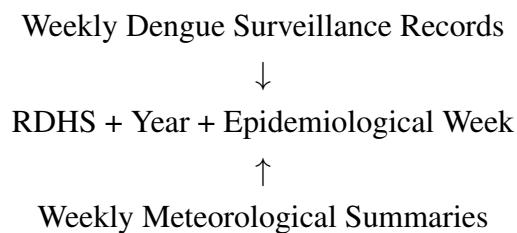
A.3 Spatiotemporal Dataset Alignment

The epidemiological and meteorological datasets originate from different sources and are therefore standardised before integration. The primary spatiotemporal matching variables are:

- RDHS administrative health area;
- calendar year; and
- epidemiological week.

These variables form the composite key used to associate dengue incidence with the corresponding climatic observations.

Conceptually, the integration process is:



The resulting structure is a longitudinal RDHS-week panel dataset in which an observation represents a particular RDHS area during a particular epidemiological week.

A.4 Data Quality and Missing-Value Handling

The historical Weekly Epidemiological Reports contain structural and formatting differences across years. Automated extraction was therefore not sufficiently reliable for all historical documents. More than 540 weekly reports were reviewed during construction of the epidemiological dataset, and the required values were manually transferred into a standardised structure.

During preprocessing, RDHS names, year identifiers, and epidemiological-week values were normalised to support consistent integration with the meteorological dataset.

Missing dengue values are not automatically interpreted as zero because an unreported value and a genuine zero case count represent different conditions. The intermediate data-preparation workflow applies within-RDHS linear interpolation between adjacent valid observations where required to preserve the chronological structure of the panel dataset.

Because an interpolated value is not equivalent to a directly observed surveillance value, the effect of imputation must be considered during final model training and evaluation.

A.5 Data Acquisition Constraints

Two important constraints affected the final data scope.

Human-mobility data. Telecommunications-based mobility information was considered during the initial research design. However, the required data could not be obtained because of privacy, security, and institutional data-governance restrictions. Human mobility was therefore removed from the implemented predictor set.

Meteorological observations. The research initially considered official historical weather observations from the Sri Lanka Department of Meteorology. Continuous localised historical data for the complete study period and geographical coverage could not be obtained within the available undergraduate research resources. NASA POWER was therefore selected as a consistently available meteorological source for all target RDHS areas.

Appendix B

Experimental Configuration Reference

This appendix summarises the principal experimental settings defined for the predictive stage of the research. The configuration is provided as a concise reference to the modelling and evaluation methodology described in Chapter 3.

B.1 Prediction Tasks

The proposed early-warning framework contains two complementary predictive tasks.

1. **Regression task:** Estimate the expected number of dengue cases for each RDHS area four weeks ahead.
2. **Classification task:** Estimate whether an RDHS area is expected to enter an outbreak or high-risk condition four weeks ahead.

The purpose of this dual-stream design is to provide both a numerical forecast and an interpretable warning output.

B.2 Candidate Forecasting Models

Three representative modelling approaches are selected for comparison.

Table B.1: Roles of the candidate forecasting approaches

Model	Approach	Role in the Research
SARIMAX	Statistical time-series model	Provides the statistical baseline while representing temporal dependence, seasonality, and external climatic predictors.
LightGBM	Machine-learning model	Evaluates nonlinear relationships among structured historical, climatic, lagged, seasonal, and geographical features.
LSTM	Deep-learning sequence model	Evaluates the performance of sequential learning from historical dengue and climatic observations for four-week-ahead prediction.

The models will be compared under a common dataset, forecast horizon, and time-aware evaluation procedure. No model is assumed to be superior before the comparative experiments are completed.

B.3 Climate-Lag Configuration

The research does not assume that climatic conditions affect dengue incidence during the same week or that a single fixed lag applies to all RDHS areas.

Candidate lag periods from one to twelve weeks will be investigated for:

- precipitation;
- atmospheric temperature; and
- relative humidity.

For a climatic variable observed at week t , the candidate historical features can therefore be represented conceptually as:

$$X_{t-1}, X_{t-2}, X_{t-3}, \dots, X_{t-12}$$

The purpose of this analysis is to determine whether useful climate-to-dengue relationships occur after different delays across RDHS areas and climatic variables.

The nationally aggregated exploratory figures presented in Chapter 4 are not used to assign a precise local lag. RDHS-specific lag values must be determined using the original local weekly observations.

B.4 Time-Aware Validation Configuration

Walk-forward validation is selected as the primary evaluation strategy because the forecasting system must predict future observations using only information available from previous periods.

An illustrative expanding-window validation sequence is shown in Table B.2.

Table B.2: Illustrative walk-forward validation sequence

Training Period	Testing Period
2016–2020	2021
2016–2021	2022
2016–2022	2023
2016–2023	2024

The sequence is illustrative of the time-aware experimental design described in Chapter 3.

B.5 Evaluation Metrics

Regression and classification outputs will be evaluated separately.

Regression metrics:

- Mean Absolute Error (MAE);
- Root Mean Squared Error (RMSE);
- Mean Absolute Percentage Error (MAPE); and
- Forecast Bias.

Classification metrics:

- Precision;
- Recall;
- F1-score; and
- ROC-AUC.

Recall receives particular attention in the outbreak-alert stream because a missed high-risk period may have greater public-health consequences than an unnecessary warning.

B.6 Intermediate-Stage Implementation Status

Table B.3 summarises the status of the principal research components at the intermediate reporting stage.

Table B.3: Implementation status at the intermediate research stage

Research Component	Status
Historical WER dengue-data preparation	Completed
NASA POWER meteorological-data retrieval workflow	Completed
Daily-to-weekly meteorological aggregation	Completed
RDHS-week epidemiological and climatic data integration	Completed
Exploratory data analysis	Completed
Formal RDHS-specific 1–12 week climate-lag analysis	To be completed
Comparative SARIMAX, LightGBM, and LSTM evaluation	To be completed
Four-week-ahead outbreak-classification evaluation	To be completed
Complete dashboard integration	To be completed
Formal dashboard usability evaluation	To be completed

The status table is included to distinguish clearly between components that have already been implemented and those that remain part of the subsequent experimental and system-development stages.